If the density of water increases in the sea, it indicates that there is an increase in salinity i.e., with the increase in ocean salinity, its density also increases.
The distribution of salinity is divided into two types namely horizontal and vertical. No specific law has been formulated yet regarding the vertical distribution of salinity in the ocean. At some places, salinity increases with an increase in depth while at her places it decreases with an increase in depth.

10. Where is Great Salt Lake located ?
(a) Iran (b) U.S.A.
(c) India (d) Turkey
Uttarakhand U.D.A./L.D.A. (Mains) 2006, 2007

Ans. (b)

The Great Salt Lake is located in Utah province of the United States of America. The salinity of this lake is 220‰.

11. The average salinity of water of Arabian Sea is-
(a) 25 ppt (b) 35 ppt
(c) 45 ppt (d) 55 ppt
53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

Between 0°-10° latitudes in the Indian Ocean, the salinity is 35.14‰ , but in the Bay of Bengal, it decreases to 30‰. At the same time salinity of the Arabian Sea is found to be 36‰ because of comparatively dry weather, more vaporization takes place and water brought by rivers is also less. Therefore, the nearest option (b) is correct.

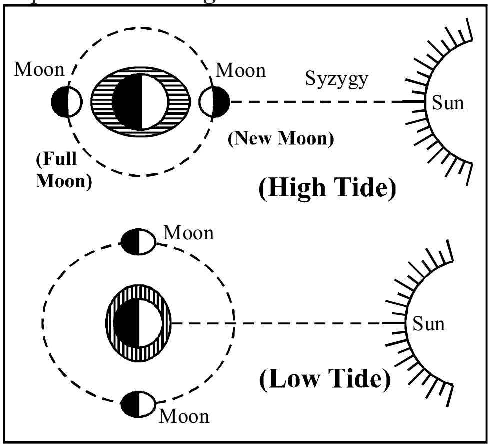
12. Which of the following seas has the highest average salinity?
(a) Black Sea (b) Yellow Sea
(c) Mediterranean Sea (d) Dead Sea
(e) None of the above/More than one of the above
60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

Among the given options Dead Sea has the highest average salinity (238‰). Assal Lake has the highest average salinity in the world i.e. 348‰.

High Tide, Low Tide

*Tides are the rise and fall of sea levels caused by the combined effects of the gravitational forces exerted by the Moon and the Sun, and the rotation of the Earth. The rotation of the Earth creates two forces - **the centripetal and centrifugal forces**, which play a major role in the formation of tides. This means that the side of the Earth that faces the Noon experiences a gravitational force, greater than that of the Earth itself. Hence we experience high tides. On the other hand the side of Earth that is facing away from moon experiences less gravitational force from the Moon, as a result, we experience high (indirect high tides) tides. Between these two high tides water level goes down this is called low tides (ebb). Thus in the time span of 24 hrs. and 50 minutes we experience two **High Tides** and two **Low Tides**.



*When the Sun, the Moon and the Earth are aligned in a straight line, we experience a Spring Tide. In astronomy, this situation is known as **syzygy**. It occurs on a full moon or new moon day. On the other hand, when the Sun, the Moon and the Earth are at a right angle to each other the gravitational forces of these celestial bodies act against each other resulting in Neap Tide. Neap Tide is observed on the 7th or 8th day of each lunar month (14 days appx.) During Neap Tides, there is a decrease of 20% in the Spring Tides whereas the Neap Tides are higher than the normal occurrences. *The Noon is the closest celestial body to Earth hence its gravity has a significant impact on our planet. The moon is responsible for the sea waves of the oceans. *The effect of Moon on the tidal forces of our planet is 2.17 times greater than that of the Sun. *The city of Southampton in England experiences 4 tides every day. This occurs because two tides travel from English Channel and two from the North Sea at different intervals.

1. What are the causes of high-tide low tide formation in the oceans ?
(a) Due to the effect of the sun
(b) Due to the rotation of the Earth

(c) Due to the combined effect of sun and the moon
(d) Due to Gravitation, Centripetal force and the centrifugal force

U.P.P.C.S. (Pre) 1991

Ans. (d)

High tide/low tide occur as a result of the gravitational pull between the Earth and the Moon alongwith the two forces namely centripetal force and centrifugal force acting upon the Earth. Ocean water is kept at equal levels around the planet by the Earth's gravity pulling inward, but Moon's gravitational force is strong enough to disrupt this balance by accelerating the water in the hemisphere that faces the Moon. When the Moon's force of attraction is stronger than the centrifugal force of the Earth, causes high tides to occur. On the other hand, when the centrifugal force is stronger than the Moon's force of attraction indirect (high tides) occur. Between these two high tides water level decreases which is called low tides (ebb). Earth experiences high and low tides twice in 24 hours and 50 minutes.

2. Tides occur in oceans and seas due to which among the following?
1. Gravitational force of Sun
2. Gravitational force of Moon
3. Centrifugal force of Earth
Select the correct answer using the codes given below:

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2015

Ans. (d)

See the explanation of the above question.

3. The cause of producing indirect high tide is –

(a) Gravitational force of the Moon
(b) Gravitational force of the Sun
(c) Centrifugal force of the Earth
(d) Gravitational force of the Earth

R.A.S./R.T.S. (Pre) 1999

Ans. (c)

See the explanation of the above question.

4. Statement (A) : During the times of neap-tide, high-tide is below normal and low-tide is above normal.
Reason (R) : Neap-tides occurs during the new-moon instead of full-Moon.

Choose the correct answer using following options –

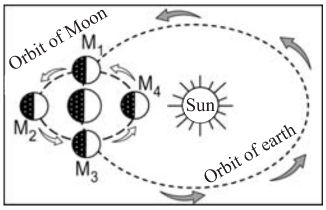
(a) (A) and (R) both are correct, and (R) is correct explanation of (A).
(b) (A) and (R) both are correct, and (R) is not correct explanation of (A).
(c) (A) is correct, but (R) is wrong.
(d) (A) is wrong, but (R) is correct.

I.A.S. (Pre) 1998

Ans. (c)

During neap-tides, high-tide remains 20% less than normal and low-tide at this time remains higher than normal. Neap-tide opposite to spring-tide doesn't occur at new-moon or full-moon but on 7th or 8th days from these two extremes. Therefore, statement (A) is correct but the reason (R) is wrong.

5. In which one of the following positions given in the figure sea-tide would have maximum height ?



(a) M_1 (b) M_2
(c) M_3 (d) M_4

I.A.S. (Pre) 1999

Ans. (d)

In the figure, M_2 is the position of Full Moon and M_4 is the position of New Moon. In both of these positions the Sun the Moon and the Earth remain in a linear position and this situation is known as 'Syzygy'. In both of these situations, spring tides occur. Perigean spring tide occur at New Moon or Full Moon when the moon is closest to Earth. During New Moon's (closest) Perigean spring tides occur.

6. Spring Tide occurs :

(a) When the Sun the Earth and Moon are in a straight line
(b) When the Sun and Moon make a right angle
(c) When a strong wind blows
(d) When the night is very cold

U.P.P.C.S. (Pre) 1999

Ans. (a)

See the explanation of the above question.

7. The high tide in the Ocean is caused by

(a) Earthquake (b) Sun